

**Math 2551 Worksheet Answers - Review from Single-Variable
Calculus and Linear Algebra**

1. $2\mathbf{v} = \langle 2, 4, 2 \rangle$, $\mathbf{v} - \mathbf{w} = \langle 2, 2, -1 \rangle$, $\mathbf{v} \cdot \mathbf{w} = 1$.
2. $a + 15b + 7c$
3. $\mathbf{u} \cdot \mathbf{v} = 0$ (because the vectors are orthogonal)
4. (a) $\mathbf{v} \cdot \mathbf{u} = -25$
(b) $\cos(\theta) = \frac{-25}{25} = -1$
(c) $(3\mathbf{v}) \cdot (2\mathbf{u}) = 3(2)(-25) = -150$.
5. (a) $16/15$
(b) $\frac{\pi}{2}$
(c) $\ln(4) - \frac{3}{4}$
6. $f_{\max} = 2e^4$, since the function has no critical points and is increasing on this interval.
7. Area = $\frac{9}{2}$.

Math 2551 Worksheet Answers: $\mathbb{R}^3$

1. $x = 0, y = 3\sqrt{3}/2, z = 3/2$.
2. The first car is faster, at approx 43.28 km/h.
3. $(x - 1)^2 + (y + 2)^2 + (z - 3)^2 = 16$.

Math 2551 Worksheet 2 Answers: $\mathbb{R}^3$ and Cross Products

1. Let $P = (1, -1, 2)$, $Q = (2, 0, -1)$, and $R = (0, 2, 1)$.
 - (a) $2\sqrt{6}$
 - (b) $\frac{1}{\sqrt{6}}\langle 2, 1, 1 \rangle$
2.
 - (a) True
 - (b) True
 - (c) True
3.
 - (a) Makes sense, scalar
 - (b) Does not make sense
 - (c) Makes sense, vector
 - (d) Does not make sense
4. Max value is $\sqrt{13}$, given by $\mathbf{v} = \mathbf{u}/|\mathbf{u}|$ and min value is $-\sqrt{13}$, given by $\mathbf{v} = -\mathbf{u}/|\mathbf{u}|$.

Math 2551 Worksheet Answers: Lines and Planes

1. (a) $\mathbf{r}(t) = \langle 1 - 2t, 2 - 2t, -1 + 2t \rangle$
(b) $\mathbf{r}(t) = \langle t, -7 + 2t, 2t \rangle$
(c) $\mathbf{r}(t) = \langle 1 + 14t, 2t, 15t \rangle$
2. vector: $3\mathbf{i} - 3\mathbf{j} + 3\mathbf{k}$.
plane: $3(x - 2) - 3(y - 1) + 3(z + 1) = 0$
3. many correct solutions; pick two distinct points off of the line PQ which are not collinear with the given points. Two such planes are $-2(x - 1) + 3(y + 1) + 5(z - 1) = 0$ and $-3(x - 1) + 3(y + 1) + 3(z - 1) = 0$.
4. $(2, 7, 3)$
5. Parallel: $\langle A_1, B_1, C_1 \rangle = \lambda \langle A_2, B_2, C_2 \rangle$ for some $\lambda \neq 0$
Perpendicular: $\langle A_1, B_1, C_1 \rangle \cdot \langle A_2, B_2, C_2 \rangle = 0$
6. $(4, 1, 3)$

Math 2551 Worksheet Answers: Quadric Surfaces

1. (a) elliptical cylinder, oriented along the y -axis, cross-sections are ellipses in the $y = k$ planes or vertical/horizontal lines in the $x = k$ and $z = k$ planes.
- (b) ellipsoid, centered at the origin, wider in the z and y directions than the x direction, cross-sections are circles in the $x = k$ planes (if $k < 1$), ellipses in the $z = k$ and $y = k$ planes ($k < 3$)
- (c) circular cone, oriented along the y -axis, cross sections are circles in the $y = k$ planes and lines in the $x = k$ and $z = k$ planes
- (d) elliptical paraboloid, oriented in the positive z direction, shifted up 4 units, cross-sections are circles in the $z = k$ planes for $k \geq 4$ and parabolas in the $x = k$ or $y = k$ planes
- (e) sphere, centered at $(0, 0, 0)$, radius 4, cross-sections are circles in $x = k$, $y = k$, and $z = k$ planes for $0 \leq k \leq 4$.

Math 2551 Worksheet Answers: Curves in Space and Their Tangents

1. This curve's graph is a spiral, narrowing to a point at the origin when $t = 0$ and widening outward around the z -axis for larger/smaller t .
2. $\mathbf{r}(t) = \langle 3 \cos(t), 3 \sin(t), 2 - 3 \sin(t) \rangle, 0 \leq t \leq 2\pi$.
3. All three functions describe part of the same set of points in $\mathbb{R}^3$, which lie above the line $y = x$ in the xy -plane and form a parabola in the plane $x = y$. $\mathbf{f}$ traces out all of the points on this parabola, $\mathbf{g}$ only those in the first octant, and $\mathbf{h}$ only those which lie above the square $[-1, 1] \times [-1, 1]$.
4. Many possible answers; depending on the domain and functions involved. If the domain is bounded, e.g. $[a, b]$, then letting $s = b + (a - b)t$ and taking $\mathbf{r}(s)$ as the new parametric equations works. If the domain is $(-\infty, \infty)$, we can just let $s = -t$.
5. (a) $y = \frac{2}{9}x^2$ for $x > 0$
 (b) $\mathbf{v}(\ln(3)) = 3\mathbf{i} + 4\mathbf{j}$
 $\mathbf{a}(\ln(3)) = 3\mathbf{i} + 8\mathbf{j}$.
6. $x(s) = s, y(s) = \frac{s}{3}, z(s) = s$
7. $(1, 1, 1)$ (where the first parameter is 1 and the second is 0). The angle is $\arccos(3/\sqrt{14})$. The bees would not collide, since the first bee reaches the point at $t = 1$ and the second bee at $t = 0$.
8. $-\frac{1}{\sqrt{2}}(x - \frac{1}{\sqrt{2}}) + \frac{1}{\sqrt{2}}(y - \frac{1}{\sqrt{2}}) + 6(z - 0) = 0$
 OR $x - y - 6\sqrt{2}z = 0$

Math 2551 Worksheet 5 Answers: Calculus of Vector-Valued Functions

1. $\mathbf{r}(t) = \langle -\frac{1}{2}t^2 + 5t + 10, -\frac{1}{2}t^2 + 10, -\frac{1}{2}t^2 + 10 \rangle, \quad t \geq 0.$
2. A baseball is hit when it is 2.5 ft above the ground. It leaves the bat with an initial velocity of 140 ft/sec at a launch angle of 30° . At the instant the ball is hit, an instantaneous gust of wind blows against the ball, adding a component of $-14\hat{i}$ (ft/sec) to the ball's initial velocity. A 15 ft high fence lies 400 ft from the home plate in the direction of the flight. (Note that gravity, $g = 32$ ft/sec²)
 - (a) Sorry, no sketch. :)
 - (b) $\vec{r}(t) = (140 \cos 30^\circ - 14)t\hat{i} + (2.5 + (140 \sin 30^\circ)t - 16t^2)\hat{j} = (70\sqrt{3} - 14)t\hat{i} + (2.5 + 70t - 16t^2)\hat{j}.$
 - (c) $y_{\max} = \frac{(140 \sin 30^\circ)^2}{64} + 2.5 = \frac{70^2}{64} + 2.5 = 79.0625$ ft., which is reached at $t = \frac{140 \sin 30^\circ}{32} = \frac{70}{32} = 2.1875$ s.
 - (d) For the time, solve $y = 2.5 + 70t - 16t^2 = 0$ for t . Using quadratic formula, we have $t = 4.41$ s. Then, the range at $t = 4.41$ is $x(4.41) = (140 \cos 30^\circ - 14)(4.41) = 472.94$ ft.
 - (e) For the time, solve $y = 2.5 + 70t - 16t^2 = 20$ for t . Using quadratic formula, we have $t = 0.27, 4.11$ seconds. Then, the range at those times are $x(0.27) = 29$ ft and $x(4.11) = 441$ ft.
 - (f) Yes, according to part (d), the ball is still 20 feet above the ground when it is 441 feet from home plate.

Math 2551 Worksheet Answers: Arc Length

1. $\mathbf{T}(t) = \frac{1}{13} \langle 12 \cos(2t), -12 \sin(2t), 5 \rangle$

length: 13π

2. $(0, 5, 24\pi)$

3. $\int_0^1 \sqrt{\left(\frac{2}{t+1}\right)^2 + (2e^{2t} + 1)^2 + (2 \sin(t) \cos(t))^2} dt$

4. $\sqrt{6}$

Math 2551 Worksheet Answers: Curvature and Normals

1. $\mathbf{T}(t) = \frac{1}{1 + e^{2t}} \langle \sqrt{2}e^t, e^{2t}, -1 \rangle$

$$\mathbf{N}(t) = \frac{1}{1 + e^{2t}} \langle 1 - e^{2t}, \sqrt{2}e^t, \sqrt{2}e^t \rangle$$

$$\kappa(t) = \frac{\sqrt{2}e^{2t}}{(e^{2t} + 1)^2}$$

2. $\mathbf{T}(t) = \cos(t)\mathbf{i} + \sin(t)\mathbf{j}$

$$\mathbf{N}(t) = -\sin(t)\mathbf{i} + \cos(t)\mathbf{j}$$

$$\kappa(t) = \frac{1}{t}$$

3. $\mathbf{T} = \left\langle \frac{1}{\sqrt{1+t^4}}, \frac{t^2}{\sqrt{1+t^4}} \right\rangle$

$$\mathbf{N} = \left\langle \frac{-t^2}{\sqrt{1+t^4}}, \frac{1}{\sqrt{1+t^4}} \right\rangle \text{ if } t > 0 \text{ and } \left\langle \frac{t^2}{\sqrt{1+t^4}}, \frac{-1}{\sqrt{1+t^4}} \right\rangle \text{ if } t < 0.$$

The normal vector does not exist when $t = 0$; as t passes from negative to positive values the normal vector changes which side of the curve it is on.

4. The point of greatest curvature occurs at $x = 0$. Using the parameterization $\mathbf{r}(t) = \langle t, t^2 \rangle$ gives $\kappa(t) = \frac{2}{(1 + 4t^2)^{3/2}}$, which is maximized when $t = 0$.

Math 2551 Worksheet Answers: Multivariable Functions

1. Find and sketch the domain for each function.
 - (a) $\{(x, y) \mid x - y \geq 1\}$
 - (b) $\{(x, y) \mid x \geq 4, |y| \geq 1\} \cup \{(x, y) \mid x < 4, |y| < 1\}$
 - (c) $\{(x, y) \mid 4x^2 - 1 \leq y \leq 4x^2 + 1\}$
 - (d) All of $\mathbb{R}^2$ except the circle $x^2 + y^2 = 4$
 - (e) All of the disk $x^2 + y^2 < 4$ except the circle $x^2 + y^2 = 3$.
2.
 - (a) a collection of concentric ellipses
 - (b) a collection of unequally spaced concentric circles
 - (c) a collection of unequally spaced parallel lines
 - (d) a collection of equally spaced parallel lines
 - (e) a collection of unequally spaced concentric circles
 - (f) a collection of equally spaced concentric circles
 - (g) two straight lines and a collection of hyperbolas
3. The plane $4x + y + 3 = 0$, except for those points with $x + y = -1$.
4. The sphere $3 = x^2 + y^2 + z^2$
5. When $x = 0$, the cross-section is the parabola $z = y^2$.
When $y = 0$, the cross-section is the parabola $z = x^2$.
When $x = y$, the cross-section is the line $z = 0$.
The level curves are pairs of parallel lines $y = x \pm \sqrt{k}$.

Math 2551 Worksheet Answers: Limits and Continuity

1. $\frac{1}{36}$
2. $\frac{1}{12}$
3. $\frac{1}{4}$
4. Does not exist.
5. f is continuous on its entire domain: all (x, y) such that neither $x = 0$ nor $y = 0$.
6. f is continuous on its entire domain: all (x, y, z) except the sphere $x^2 + y^2 + z^2 = e$.

Math 2551 Worksheet 8 Answers - Review for Exam 1

1. $\int_0^1 \sqrt{1 + e^{2t}} dt$
2. $\ell(s) = \langle 0, -s, -s \rangle$
3. $\mathbf{T}(t) = \langle -\sin(t), \cos(t) \rangle$
 $\mathbf{N}(t) = \langle -\cos(t), -\sin(t) \rangle$
Into
4. $\kappa = \frac{a}{1 + a^2}$
5. $-3(x - 3) + (y - 2) - 2z = 0$
Intersection point is $(0, 3, -2)$, when $t = 1$.
6. (a) $\{(x, y) \mid |x| \leq 3, |y| \geq 2\}$
(b) $1 \leq x^2 + y^2 \leq 3$
(c) $\frac{x^2}{16} + \frac{y^2}{4} \leq 1$
7. $\mathbf{r}(t) = t^2\mathbf{i} + t^3\mathbf{j} + \frac{2}{3}(t^{3/2} - 1)\mathbf{k}$
8. No, because the limit of f as $(x, y) \rightarrow (0, 0)$ does not exist.

Math 2551 Worksheet Answers: Partial Derivatives

1. (a) $f_x(-2, -1) \approx 0.75$
 (b) $f_y(-2, -1) \approx 1.5$
 (c) There are several possible points (these are places where the tangent to a contour is horizontal): $(0, -.5), (-.5, -1.25)$, etc.
 (d) Again, there are many possible points; any point on the 4, 5, 6 contours in quadrant IV will work.
2. (a) C_T : (meters/second)/ degree Celsius - this gives the change in speed for each one degree C of temperature increase. C_S (meters/second)/(grams/liter) - this gives the change in speed for each one gram/liter increase in salinity C_D : (meters/second)/meter - this gives the change in speed for each one meter increase in depth below the surface
 (b) $C_T = 4.5 - 0.1T + 0.0009T^2 - 0.01(S - 35)$ $C_S = 1.5 - 0.01T$ $C_D = 0.015$
 (c) At $(T, S, D) = (10, 35, 100)$, we have $C_T = 3.59, C_S = 1.4, C_D = 0.015$. This tells us that if we increase the temperature, salinity, or depth from these conditions the speed of sound will increase as well.
3. $f_{xx} = e^x, f_{xy} = f_{yx} = \frac{1}{y}, f_{yy} = -\frac{x}{y^2}$
4. $f_x = \sin(yz), f_y = xz \cos(yz), f_z = xy \cos(yz), f_{xzz} = -y^2 \sin(yz)$
5. Find the total derivative Df at the given point for each function below. Remember that Df is the matrix of (partial) derivatives of the function and if f is a function from $\mathbb{R}^n$ to $\mathbb{R}^m$ then Df is a $m \times n$ matrix.
 - (a) $Df(2) = f'(2) = [24]$
 - (b) $D\mathbf{f}(\pi/2) = \mathbf{f}'(\pi/2) = \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix}$
 - (c) $Df(1, 2) = [-1/2 \quad 1/2]$
 - (d) $Df(1, 2, 3) = [-e^3 \quad 2e^3 \quad 6]$
 - (e) $D\mathbf{f}(1, 1) = \begin{bmatrix} 2 & 3 \\ -1 & 1 \end{bmatrix}$
6. Note this does not have a definitive right answer - some differences may arise and that's good! Discuss!
 - (a) First y since $\partial^3 f / \partial y^3 = 0$ and the y -partial derivatives are easier
 - (b) First y , since $\partial^3 f / \partial y^3 = 0$
 - (c) First y , since $\partial^2 f / \partial y^2 = 0$
 - (d) First x , since $\partial^2 f / \partial x^2 = 0$ and the x -partial derivatives are easier.

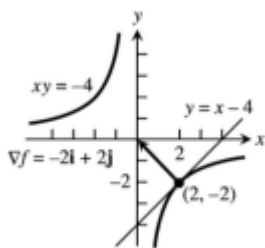
A common theme is to work with the variable with lower powers/simpler expressions first when taking mixed partials.

Math 2551 Worksheet Answers: Chain Rule

1. $\frac{dz}{dt}(t_0) = 1$
2. $t = -4/3$
3. $\frac{\partial w}{\partial r}(2, \pi/2) = 2\pi$ $\frac{\partial w}{\partial \theta}(2, \pi/2) = -2\pi$
4. $\frac{dw}{dt}(2) = -10$. $\frac{dw}{dt}(1)$ cannot be computed from the given information because we do not know the values of g_x or g_y at $(x(1), y(1)) = (1, 3)$. We do not use the values of $g(1, 0), g(-1, 2), g_x(-1, 2), g_y(-1, 2)$.

Math 2551 Worksheet Answers: Gradient and Directional Derivatives

1. (a) Negative
(b) Negative
(c) Approximately zero
(d) Positive
(e) Positive
2. Tangent line: $-2(x - 2) + 2(y + 2) = 0$



3. $D_{\mathbf{u}}g(1, -1) = \frac{21}{13}$
4. (a) Ascend at a rate of 0.8 vertical meters per horizontal meter
(b) Descend at a rate of $\sqrt{2}/10$ vertical meters per horizontal meter
(c) $\langle -0.6, -0.8 \rangle$ is the direction of largest slope with rate of ascent 1 vertical meter per horizontal meter.
5. (a) $\langle -1/\sqrt{2}, 1/\sqrt{2} \rangle$
(b) $2\sqrt{2}$
(c) $\langle 1/\sqrt{2}, -1/\sqrt{2} \rangle$
(d) $\langle 1/\sqrt{2}, 1/\sqrt{2} \rangle$

Math 2551 Worksheet Answers: Linearization and Tangent Planes, Optimization

1. $L(x, y) = e^3 - e^3(x - 1) + 2e^3(y - 2)$
2. $L(x, y, z) = z$
3. $f(2.95, 7.1) = 4 - 1/40$
4. No, there are two different tangent planes. No, the function is not differentiable.
5. $z = 1 - 1/2(x - 1) + 1/2(y - 2)$

Math 2551 Worksheet Answers: Optimization I

1. Saddle point at $(0, 0)$ and local minimum at $(0, -2)$.
2. Saddle point at $(0, 0)$ and local maximum at $(-1, -1)$.
3. Yes, this must be a saddle point because $f_{xx}(a, b)f_{yy}(a, b) < 0$ so $\det(Hf) = f_{xx}(a, b)f_{yy}(a, b) - f_{xy}^2(a, b) < 0$.
4.
 - (a) Minimum is 0 at $(0, 0)$ since $f(x, y) > 0$ for all other (x, y) .
 - (b) Maximum is 1 at $(0, 0)$ since $f(x, y) < 1$ for all other (x, y) .
 - (c) Neither since $f(x, y) < 0$ for $x < 0$ and $f(x, y) > 0$ for $x > 0$.
 - (d) Neither since $f(x, y) < 0$ for $x < 0$ and $f(x, y) > 0$ for $x > 0$.
 - (e) Neither since $f(x, y) < 0$ for $x < 0$ and $y > 0$, but $f(x, y) > 0$ for $x > 0$ and $y > 0$.
 - (f) Minimum is 0 at $(0, 0)$ since $f(x, y) > 0$ for all other (x, y) .

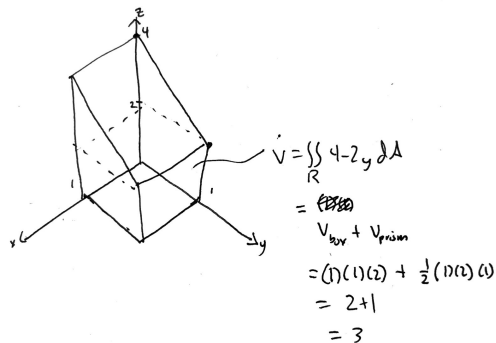
Math 2551 Worksheet Answers: Optimization II

1. The absolute maximum is 17, achieved at $(0, 4)$ and $(4, 4)$, and the absolute minimum is 1, achieved at $(0, 0)$.
2. The dimensions are $3 \times 3 \times 3$ and the surface area is 54.

Math 2551 Worksheet Answers: Lagrange Multipliers

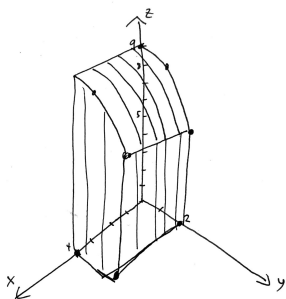
1. $V(2, 2, 1) = 4$ cubic units
2. The extreme values are 1 and 2.
3. The extreme values are 1 and $\sqrt{3}$.
4. The closest point is $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$ and the farthest point is $(-1, -1, 2)$.
5. $\frac{8r^3}{3\sqrt{3}}$

Math 2551 Worksheet Answers: Double Integrals on Rectangles



1.

= 3



2.

3. $9 \ln(2)$ 4. $160/3$ cubic units.5. $e^3 - 4$

6. The integrals evaluate to $\pi/4$ and $-\pi/4$ respectively. This does not violate Fubini's theorem because this function is not continuous on $[0, 1] \times [0, 1]$ (it has an asymptote at $(0, 0)$)

Math 2551 Worksheet Answers: Double Integrals on General Regions

1. $2 + \frac{1}{2}\pi^2$
2. (a) Negative
(b) Positive
(c) Zero
(d) Zero
(e) Zero
3. (a) $\int_0^{\ln(3)} \int_{e^{-x}}^1 dy \, dx$ and $\int_{1/3}^1 \int_{-\ln(y)}^{\ln(3)} dx \, dy$
(b) $\int_{-1}^2 \int_{x^2}^{x+2} dy \, dx$ and $\int_0^1 \int_{-\sqrt{y}}^{\sqrt{y}} dx \, dy + \int_1^4 \int_{y-2}^{\sqrt{y}} dx \, dy$
4. $\int_0^{\sqrt{\pi}} \int_0^x \cos(x^2) \, dy \, dx = 0$
5. $\frac{2}{15}$
6. $16/3$
7. $32/3$

Math 2551 Worksheet 16 Answers: Applications, Polar Double Integrals

1. Answers will vary a bit through the estimation process

4 subdivisions: $31.75 \leq T_{avg} \leq 52.5$

16 subdivisions: $33.18 \leq T_{avg} \leq 50.06$

25 subdivisions: $36.32 \leq T_{avg} \leq 49.8$

Colorado is a rectangle, which makes it easy to subdivide. Wyoming would also work well.

2. On square: $f_{avg} = \frac{1}{1} \cdot \frac{1}{4} = \frac{1}{4}$

On quarter circle: $f_{avg} = \frac{1}{\pi/4} \cdot \frac{1}{8} = \frac{1}{2\pi}$

3. 50 people

Math 2551 Worksheet Answers: Polar Double Integrals

1. $5\pi - 26$

2. $\frac{\pi}{2}(1 - e^{-4}).$

3. $\frac{3\pi}{2} - 4.$

4. 256π

5. (a) $I^2 = \lim_{R \rightarrow \infty} \int_0^2 \pi \int_0^R e^{-r^2} r \, dr \, d\theta$

(b) $I^2 = \frac{\pi}{4}, \text{ so } I = \frac{\sqrt{\pi}}{2}$

Math 2551 Worksheet Answers: Exam 2 Review

1. All are true except b).
2. -1
3. ± 1
4. Saddle at $(0, 0)$ with $f(0, 0) = 0$, local min at $(0, 2)$ of -4 , local max at $(-2, 0)$ of 4 , saddle at $(-2, 2)$ with $f(-2, 2) = 0$
5. $(-1/2, 1/2, 1/2)$ and $(0, 1, 0)$
6. $\sin(4)$
7. No, this is less than $f(x, y)$ at all points, so it cannot possibly be the average value.
8. 1800π cubic feet

Math 2551 Worksheet Answers: Triple Integrals

1. $\frac{-58}{3}$, region of integration the rectangular prism $[-1, 1] \times [0, 4] \times [0, 1]$ or $-1 \leq x \leq 1, 0 \leq y \leq 4, 0 \leq z \leq 1$.
2. Various possibilities depending on order of integration. E.g

$$\int_0^1 \int_0^{1-x} \int_0^{1-x-y} z \, dz \, dy \, dx$$

$$3. \quad (a) \int_0^1 \int_0^1 \int_{-1}^{-\sqrt{z}} dy \, dz \, dx$$

$$(b) \int_0^1 \int_0^1 \int_{-1}^{-\sqrt{z}} dy \, dx \, dz$$

$$(c) \int_0^1 \int_{-1}^{-\sqrt{z}} \int_0^1 dx \, dy \, dz$$

$$(d) \int_{-1}^0 \int_0^{y^2} \int_0^1 dx \, dz \, dy$$

$$(e) \int_{-1}^0 \int_0^1 \int_0^{y^2} dz \, dx \, dy$$

$$4. \int_{-1}^1 \int_{1-\sqrt{1-x^2}}^{1+\sqrt{1-x^2}} \int_{x^2+y^2}^{2y} dz \, dy \, dx.$$

$$\int_0^2 \int_{y^2}^{2y} \int_{-\sqrt{z-y^2}}^{\sqrt{z-y^2}} dx \, dz \, dy.$$

Yes, all orders of integration result in a single iterated integral.

Math 2551 Worksheet Answers: Mass and Moments

1. $(\bar{x}, \bar{y}) = \left(0, \frac{9}{14}\right)$

2. Mass: $\int_{-2}^2 \int_{-\sqrt{1-x^2/4}}^{\sqrt{1-x^2/4}} \int_0^{2-x} 2 \, dz \, dy \, dx$

$$M_{yz}: \int_{-2}^2 \int_{-\sqrt{1-x^2/4}}^{\sqrt{1-x^2/4}} \int_0^{2-x} 2x \, dz \, dy \, dx$$

$$M_{xz}: \int_{-2}^2 \int_{-\sqrt{1-x^2/4}}^{\sqrt{1-x^2/4}} \int_0^{2-x} 2y \, dz \, dy \, dx$$

$$M_{xy}: \int_{-2}^2 \int_{-\sqrt{1-x^2/4}}^{\sqrt{1-x^2/4}} \int_0^{2-x} 2z \, dz \, dy \, dx$$

Math 2551 Worksheet Answers: Triple Integrals in Cylindrical & Spherical Coordinates

1. Integral: $\int_{-\pi/2}^{\pi/2} \int_0^1 \int_0^{r \cos(\theta)} r^3 \, dz \, dr \, d\theta$

Answer: $\frac{2}{5}$.

2. Integral: $\int_0^{\pi/6} \int_0^{\pi/2} \int_0^a \rho^2 \sin(\phi) \, d\rho \, d\phi \, d\theta$

Answer: $\frac{a^3 \pi}{18}$.

3. Integral: $\int_0^{2\pi} \int_{\pi/4}^{\pi/2} \int_0^2 \rho^2 \sin(\phi) \, d\rho \, d\phi \, d\theta$

Answer: $\frac{8\sqrt{2}\pi}{3}$

4. Integral: $\int_0^{2\pi} \int_1^3 \int_0^{9-r^2} r \, dz \, dr \, d\theta$

Answer: 32π

5. Integral: $2 \left(\int_0^{2\pi} \int_{\pi/6}^{\pi/4} \int_{\csc(\phi)}^2 \rho^2 \sin(\phi) \, d\rho \, d\phi \, d\theta + \int_0^{2\pi} \int_{\pi/4}^{\pi/2} \int_{\sqrt{2}}^2 \rho^2 \sin(\phi) \, d\rho \, d\phi \, d\theta \right)$

Answer: $\frac{12\sqrt{3}-4}{3}\pi$.

6. (a) Integral: $\int_0^\pi \int_0^{2\sin(\theta)} \int_0^{4-r\sin(\theta)} r \, dz \, dr \, d\theta$

Answer: 3π .

(b) $\bar{x} = 0$.

Math 2551 Worksheet Answers: Change of Variables

1. $\sin^2(\theta) - \cos^2(\theta) = -\cos(2\theta)$

2. The elliptical region $\frac{x^2}{a^2} + \frac{y^2}{b^2} \leq 1$.

Many possible answers: any such should involve an integration problem where the region of integration is an ellipse with major and minor axes parallel to the x and y -axes.

3. $x = \sqrt{\frac{u}{v}}, y = \sqrt{uv}$ maps $[1, 3] \times [1, 3]$ onto R

4. $\int_0^1 \int_{-3v}^{-3v+3} -2v \, du \, dv$

5. $\ln(9)$

6. $e + \frac{1}{e}$

Math 2551 Worksheet Answers: Scalar Line Integrals

1. There are many possible correct answers! Here are some.

(a) $\mathbf{r}(t) = \langle 3t, -2t + 1, 4t - 2 \rangle, \quad 0 \leq t \leq 1$

(b) $\mathbf{r}(t) = \langle -3t + 3, 2t - 1, -4t + 2 \rangle, \quad 0 \leq t \leq 1$

(c) $\mathbf{r}(t) = \langle 3 \sin(t), 3 \cos(t) \rangle, \quad \pi \leq t \leq 3\pi$

(d) $\mathbf{r}(t) = \langle t^2, -t \rangle \quad -2 \leq t \leq 1$

2. The integral to compute is $\int_0^{2\pi} 20 \, dt$ since $f(\mathbf{r}(t)) = 4$ and $\|\mathbf{r}'(t)\| = 5$. Its value is 40π .

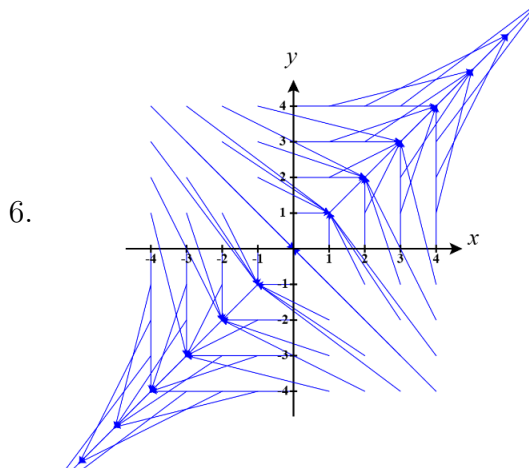
3. The integral to compute is $Ds \int_{-2}^1 4t^2 + 1 \, dt$. Its value is 15.

4. The integral to compute is $Ds \int_0^\pi e^{-t} e^t \sqrt{3} \, dt \sqrt{3}\pi$

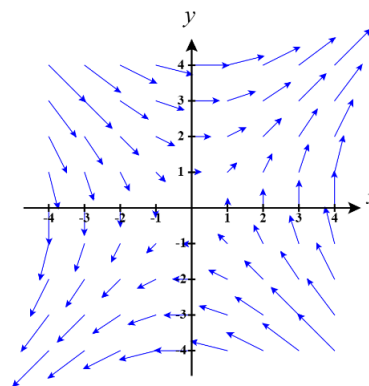
5.

(x, y)	$(-2, 0)$	$(-1, 2)$	$(0, -2)$	$(1, 1)$	$(2, 3)$	$(3, 2)$	$(-1, 0)$	$(1, 3)$
$\mathbf{F}(x, y)$	$\langle 0, -2 \rangle$	$\langle 2, -1 \rangle$	$\langle -2, 0 \rangle$	$\langle 1, 1 \rangle$	$\langle 3, 2 \rangle$	$\langle 2, 3 \rangle$	$\langle 0, -1 \rangle$	$\langle 3, 1 \rangle$

Without rescaling vectors to fit better:



With rescaling:



7. $\nabla f(x, y) = \langle x - y, y - x \rangle$

Math 2551 Worksheet Answers: Vector Line Integrals

1. Top left: 0
Bottom left: Negative
Center: Positive
Top right: 0
Bottom right: Negative
2. 1
3. $\frac{25}{6}$
4. Circulation: $\frac{32}{3}$
Flux: 0
5. Many possible examples: the top left and top right examples from 1), ∇f for any f together with a closed curve $\mathbf{r}$, any example where $\mathbf{F} \cdot \mathbf{r}'(t) = 0$, i.e. the field and curve are orthogonal, and more.

Math 2551 Worksheet Answers: Potentials and Conservative Vector Fields

1. $f(x, y) = 12x^2y + 2y^3$

2. $f(x, y, z) = x^2y - z^2y$ and $\int_C \mathbf{F} \cdot d\mathbf{r} = -16$.

3. $7 = a, 2 = c, 4 = d$, no restriction on b or e .

4. $f(x, y, z) = \frac{x}{y} + z^2 - z$.
 $\int_C \mathbf{F} \cdot d\mathbf{r} = 1/2$

5. 2π

Math 2551 Worksheet Answers: Curl, Divergence, Green's Theorem

1. $\nabla \cdot \mathbf{F}$ is 0 in all quadrants.
 $(\nabla \times \mathbf{F}) \cdot \mathbf{k}$ is negative in all quadrants.
2. $\nabla \cdot \mathbf{F} = 0$
 $(\nabla \times \mathbf{F}) \cdot \mathbf{k} = -\frac{1}{4}.$
3. (a) $\mathbf{r}(t) = \langle 3 \cos(t), 4 \sin(t) \rangle, 0 \leq t \leq 2\pi.$
(b) Circulation: 0
Flux: 48π
4. Circulation: $-44/15$
Flux: The integrand is very difficult to work with, so we should not use Green's theorem here.
5. $-1/12$
6. One answer (linear algebra!) $\mathbf{r}(s, t) = s\langle 1, -1, 0 \rangle + t\langle 0, 1, -1 \rangle = \langle s, t - s, -t \rangle, s, t \in \mathbb{R}.$

Math 2551 Worksheet Answers: Surfaces

1. One answer: $\mathbf{r}(u, v) = \langle u, 4 - u^2, v \rangle$, $-2 \leq u \leq 2, 0 \leq v \leq 2$.
2. One answer: $\mathbf{r}(\phi, \theta) = \langle 2 \sin(\phi) \cos(\theta), 2 \sin(\phi) \sin(\theta), 2 \cos(\phi) \rangle$ with $0 \leq \phi \leq \pi/6$ and $0 \leq \theta \leq 2\pi$.
SA: $8\pi \left(1 - \frac{\sqrt{3}}{2}\right)$
3. The tangent plane is $-\sqrt{2}(x - \sqrt{2}) - \sqrt{2}(y - \sqrt{2}) + 2(z - 2) = 0$. This is the cone $z = \sqrt{x^2 + y^2}$.
4. $\frac{2}{3}\pi(2^{3/2} - 1)$
5. $\frac{\pi}{6}(5^{3/2} - 1)$

Math 2551 Worksheet Answers: Surface Integrals

1. 0

2. $\frac{10\pi}{3}$

3. 0

Math 2551 Worksheet Answers: Stokes' Theorem

1. H and P have the same oriented boundary curve C provided that they are oriented in the same way, so by Stokes' theorem the given integrals must be equal.
2. 0
3. 3

Math 2551 Worksheet Answers: Review for Exam 3

1. $\int_0^{2\pi} \int_{2\pi/3}^{\pi} \int_2^5 \rho^4 \cos^2(\phi) \sin(\phi) d\rho \, d\phi \, d\theta$
2. (a) $\int_{-2}^2 \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} \int_0^{4-x^2} dz \, dy \, dx$
(b) $\int_{-2}^2 \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} \int_0^{4-x^2} dz \, dy \, dx$
3. One solution: $\int_0^1 2(t)(-t^{3/2})^2 \sqrt{1 + (\frac{3}{2}\sqrt{t})^2} \, dt.$
4. $f(x, y, z) = x^2 - \frac{1}{3}y^3 - 4 \arctan(z).$
Integral: $-\pi.$
5. $-4 \cdot \pi(2)^2$
6. $\pi a^2/4$
7. $\nabla \times \mathbf{F} = \langle 0 - 0, -(0 - 0), 0 - 0 \rangle$
8. 12π

Math 2551 Worksheet Answers: Review for Final

1. $3x + y + z = 5$. There is not a unique plane because there is not a unique normal direction perpendicular to $\langle 3, 1, 1 \rangle$.
2. $(0, 5, 24\pi)$
3. Domain $\{(x, y) \mid y \leq x^2\}$ Range $[0, \infty)$ Level curves are the parabolas $y = x^2 - c^2$ for all $c \geq 0$.
4. The limit does not exist
5. Tangent plane: $(x-1) + (y-1) - z = 0$ Linearization: $L(x, y, z) = 1 + (x-1) + (y-1) - z$
6. $\nabla f(1, 2) = \langle 2, 2 \rangle$, $Df_{\mathbf{u}}(1, 2) = 14/5$
7. $-\sin^2(1) - \sin(1)\cos(1) + \cos^2(1) + \cos(1)\cos(2) - 2\cos(1)\sin(2) - 2\sin(1)\sin(2)$
8. $(0, 1)$ saddle point, $(2, 1), (-2, 1)$ local minimum
9. min: -32 at $(2, -2)$ max: 18 at $(1, 1)$
10. min: $-1/2$ at $(1/\pm\sqrt{2}, 1/\pm\sqrt{2})$ and max: $1/2$ at $(1/\pm\sqrt{2}, 1/\mp\sqrt{2})$.
11. $\int_{-3}^3 \int_0^{\sqrt{9/4-x^2/4}} y \, dy \, dx$
12. π
13. $(\bar{x}, \bar{y}) = \left(\frac{1}{2 - \ln(4)}, \frac{1}{2 - \ln(4)} \right)$
14. $V = \frac{\pi}{6}(8\sqrt{2} - 7)$
15. $\frac{64}{5}$
16. 0
17. 3
18. $1 - e^{-2\pi}$
19. $\frac{3}{2}$
20. $\frac{208\pi}{5}$